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Double Tube-Preserving Regularization for Robust Visual Data Recovery

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1. Introduction

Tensor Robust Principal Component Analysis (TRPCA) [1] serves as a representative method for high-dimensional visual data analysis, with applications in image and video denoising [2, 3]. It aims to characterize the intrinsic low-rank structure of tensors, a property commonly found visual applications, including color images, grayscale videos, and hyperspectral images. Given a corrupted $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, TRPCA models the observation as the superposition of an underlying low-rank tensor $\mathcal{L}$ and a sparse noise tensor $\mathcal{E}$, resulting in

$$\min_{\mathcal{L},\mathcal{E} \in \mathbb{R}^{d_1 \times d_2 \times d_3}} \operatorname{rank}(\mathcal{L}) + \lambda \|\mathcal{E}\|_1 \quad \text{s.t.} \quad \mathcal{X} = \mathcal{L} + \mathcal{E}. \tag{1}$$

Here, $\lambda > 0$ is a penalty coefficient, $\operatorname{rank}(\mathcal{L})$ is the tensor rank of $\mathcal{L}$, and $\|\mathcal{E}\|_1$ quantifies the sparsity of $\mathcal{E}$ by summing the absolute values of all tensor entries. A key challenge is how to characterize the low-rank property of $\mathcal{L}$, since tensor rank does not have a unique definition.

To tackling this challenge, various tensor rank definitions of tensor rank have been proposed from different perspectives, including CANDECOMP/PARAFAC (CP) rank [4] and Tucker rank [5]. Since directly minimizing tensor rank is generally intractable, surrogate functions are commonly adopted. For example, the Sum of Nuclear Norms (SNN) [6] provides a convex relaxation of Tucker rank. Based on the tensor singular value decomposition (t-SVD) framework [4], the tensor tubal rank [7] has gained widespread interest due to its favorable algebraic properties. Owing to its convexity and ability to promote low-rank structures, the tensor nuclear norm (TNN) [7] is widely employed in TRPCA models and leads to the following formulation:

$$\min_{\mathcal{L},\mathcal{E} \in \mathbb{R}^{d_1 \times d_2 \times d_3}} \|\mathcal{L}\|_* + \lambda \|\mathcal{E}\|_1 \quad \text{s.t.} \quad \mathcal{X} = \mathcal{L} + \mathcal{E}, \tag{2}$$

where $\|\mathcal{L}\|_*$ represents the TNN of $\mathcal{L}$. Besides, the effectiveness of TNN variants has been validated in [8, 9, 11]. Since methods under the TNN framework [12, 13] have

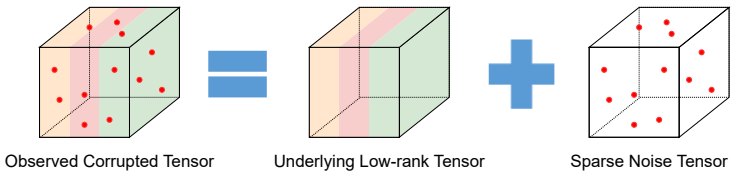


Figure 1: Illustration of the Tensor Robust Principal Component Analysis model against sparse noise.

demonstrated their effectiveness for low-rank tensor recovery, some works are committed to theoretical guarantee for these methods. Reliable theories have been proposed in [7, 14] to guarantee these TNN based methods able to result in an exact recovery, further demonstrating its effectiveness.

Apart from the low-rankness prior of tensor data, many pioneering works incorporate the smoothness prior into the TRPCA task [15, 16]. A tensor being smooth often indicates that its adjacent pixels along a tensor mode change continuously. Joint consideration of low-rankness and smoothness can more effectively capture the intrinsic structure of tensors, which has been validated in [17, 18, 20, 11]. Besides, tensor correlated total variation (t-CTV) [21] provides a unified regularization that simultaneously accounts for low-rankness and smoothness. Recently, some works devote to alleviating the curse of dimensionality and improving the efficiency of proposed methods by randomization or parallelization. Yuan et al. [22] develop efficient low-rank tensor ting approximation algorithms by using randomized full-mode projection. [23] proposes the efficient computation of low-rank tensor train approximation by several parallelizable sketching algorithms. Zhang et al. [24] propose an effective randomized algorithm via Gaussian random projection scheme under the t-SVD framework. Besides, [25] proposes two efficient low tubal rank approximation methods for high-order tensor by fusing unblocked and blocked random projection schemes.

Despite the impressive performance of these TRPCA methods, they still suffer from some limitations. Firstly, most existing TRPCA approaches employ tensor nuclear norm-based regularizers for the low-rank term $\mathcal{L}$. However, we observe that that the singular value tensor $\bar{\mathcal{S}}$ exhibits a tube sparse structure, as shown in Fig. 2. Since TNN essentially equals a scaled ℓ_1 norm of the unfolding singular value matrix (see Definition 7), it regularizes singular values individually rather than in a tube-wise man-

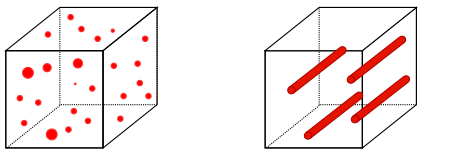


Figure 2: Illustration of two types of sparsity structures in tensors. (a) A sparse tensor whose nonzero entries are arbitrarily distributed; (b) a tube sparse tensor with four nonzero tubes.

ner. Consequently, existing TNN-based TRPCA methods fail to exploit the tube sparse structure of the singular value tensor $\bar{\mathcal{S}}$, and may thus yield suboptimal recovery performance. Additionally, existing TRPCA approaches predominantly employ the tensor ℓ_1 norm to regularize the sparse noise tensor $\mathcal{E}$. Such regularization implicitly models the corruption as element-wise sparse, with each entry treated independently. However, in many real-world scenarios, the sparse noise exhibits a tube-wise sparse structure. For example, in color images, all three channels may be corrupted at the same spatial locations, resulting in corruptions along entire tubes. Therefore, preserving and exploiting this tube sparse structure can further improve the recovery performance.

The analysis above inspires us to consider a key issue: Is it possibly to devise an enhanced TRPCA method that *simultaneously* leverages the *tube sparse structure* of both the singular value tensor of $\mathcal{L}$ and the noise tensor $\mathcal{E}$? In this paper, we provide a positive answer.

1.1. Paper Contributions

The main contributions are summarized below.

- **Model:** We propose two novel regularization terms, namely the Tube-preserving tensor nuclear norm (T2NN) and Tube-preserving Tensor ℓ_1 norm (T2L1), to exploit the tube sparse structures of the singular value tensor and tube-wise noise tensor, respectively. Based on these two regularizers, we develop a novel TRPCA model via Double Tube-preserving Regularization, termed DTR. Moreover, we extend DTR with smoothness prior and develop a novel TRPCA model with Gradient DTR (GDTR), enabling joint modeling of low-rankness and smoothness.

- **Optimization:** We devise effective optimization algorithms for both DTR and GDTR within the Alternating Direction Method of Multipliers (ADMM) framework [26]. Our algorithms can adaptively and automatically assign smaller penalization on significant singular value tubes of low-rank tensor and important tubes of the sparse tensor. Consequently, DTR provides better preservation of the underlying structures of both components simultaneously. Furthermore, theoretical convergence guarantees are established for our algorithm under the more challenging setting where both penalization variables are dynamically updated during the iterations.

- **Applications:** We apply our proposed methods for various real-world applications including color image recovery, hyperspectral image restoration, video denoising, and background modeling. Experimental results validate the effectiveness of our proposed methods.

2. Notations And Preliminaries

Initially, we first define the essential notation. We employ lowercase letters (t), bold lowercase letters ($\mathbf{t}$), bold uppercase letters ($\mathbf{T}$), and calligraphic capitals ($\mathcal{T}$) to represent scalars, vectors, matrices, and tensors, respectively. A vector $\mathbf{t} \in \mathbb{R}^d$ is denoted as $\|\mathbf{t}\|_2 = \sqrt{\sum_{i=1}^{d} t_i^2}$. For a tensor $\mathcal{T}$, the ℓ_1- and Frobenius norms take the forms $\|\mathcal{T}\|_1 = \sum_{ijk} |\mathcal{T}_{ijk}|$ and $\|\mathcal{T}\|_F = \sqrt{\sum_{ijk} \|\mathcal{T}_{ijk}\|^2}$, respectively. In the matrix case, $\|\mathbf{T}\|_* = \sum_l \sigma_l(\mathbf{T})$ represents the nuclear norm, with $\sigma_l(\mathbf{T})$ being the l-th singular value. For brevity, $[d]$ represents the integer sequence $\{1, 2, \ldots, d\}$. Key notations and acronyms are summarized in Tab. 1.

We next introduce the t-product and several related tensor definitions.

Definition 1. *(**The fold and unfold operators** [4]) Given $\mathcal{T} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, we have*

$$\text{unfold}(\mathcal{X}) = \begin{bmatrix} \mathbf{T}^{(1)} \\ \mathbf{T}^{(2)} \\ \vdots \\ \mathbf{T}^{(d_3)} \end{bmatrix}, \ \text{fold}(\text{unfold}(\mathcal{T})) = \mathcal{T}. \tag{3}$$

Notations/Acronyms	Descriptions	Notations/Acronyms	Descriptions
$\mathcal{T}(i,:,:)$	i-th horizontal slice of $\mathcal{T}$	$\mathcal{T}(:,j,:)$	j-th lateral slice of $\mathcal{T}$
$\mathcal{T}(:,:,k)$ or $\mathbf{T}^{(k)}$	k-th frontal slice of $\mathcal{T}$	$\mathcal{T}(i,j,k)$ or $\mathcal{T}_{ijk}$	(i,j,k)-th entry of $\mathcal{T}$
$\mathcal{T}^*$	transpose of $\mathcal{T}$	$\mathcal{T}_1 * \mathcal{T}_2$	tensor product of $\mathcal{T}_1$ and $\mathcal{T}_2$
DFT	Discrete Fourier Transform	$\bar{\mathcal{T}} = fft(\mathcal{T},[],3)$	DFT of $\mathcal{T}$ along the 3-rd dimension
$\vec{\sigma}_l(\bar{\mathcal{L}})$	l-th singular tube of $\mathcal{L}$	$\mathcal{E}(i,j,:)$	(i,j)-th noise tube of $\mathcal{E}$
$\|\mathcal{T}\|_*$	Tensor Nuclear Norm	$\|\mathcal{T}\|_1$	Tensor ℓ_1 Norm
$[n]$	index set $\{1,2,\cdots,n\}$	TNN	Tensor Nuclear Norm
T2NN	Tube-preserving TNN	T2L1	Tube-preserving Tensor ℓ_1 Norm

Table 1: Key acronyms and notations utilized in this work.

Definition 2. *(**The bcirc operator** [4]) For $\mathcal{T} \in \mathbb{R}^{d_1\times d_2\times d_3}$, the block circulant matrix is given by*

$$\text{bcirc}(\mathcal{T}) = \begin{bmatrix} \mathbf{T}^{(1)} & \mathbf{T}^{(d_3)} & \cdots & \mathbf{T}^{(2)} \\ \mathbf{T}^{(2)} & \mathbf{T}^{(1)} & \cdots & \mathbf{T}^{(3)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{T}^{(d_3)} & \mathbf{T}^{(2)} & \cdots & \mathbf{T}^{(1)} \end{bmatrix}. \tag{4}$$

Here, $\mathbf{T}^{(k)}$ is the k-th frontal slice of $\mathcal{T}$.

Definition 3. *(**T-product** [4]) The t-product between $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$ and $\mathcal{Y} \in \mathbb{R}^{d_2\times l\times d_3}$ produces a tensor $\mathcal{Z} = \mathcal{X} * \mathcal{Y} \in \mathbb{R}^{d_1\times l\times d_3}$, formulated as*

$$\mathcal{X} * \mathcal{Y} = \text{fold}(\text{bcirc}(\mathcal{X}) \cdot \text{unfold}(\mathcal{Y})). \tag{5}$$

Definition 4. *(**Tensor Singular Value Decomposition, T-SVD** [7]) For $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$, its t-SVD is expressed as*

$$\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*, \tag{6}$$

where $\mathcal{U} \in \mathbb{R}^{d_1\times d_1\times d_3}$, $\mathcal{V} \in \mathbb{R}^{d_2\times d_2\times d_3}$ are orthogonal, i.e., $\mathcal{U}*\mathcal{U}^* = \mathcal{U}^**\mathcal{U} = \mathcal{V}*\mathcal{V}^* = \mathcal{V}^* * \mathcal{V} = \mathcal{I}$. The tensor $\mathcal{S} \in \mathbb{R}^{d_1\times d_2\times d_3}$ is f-diagonal, meaning that all its frontal slices are diagonal matrices. The singular value tensor is defined by $\bar{\mathcal{S}} = fft(\mathcal{S},[],3)$.

Definition 5. *(**Tensor tubal rank** [7]) Given $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$, its tubal rank $rank_t(\mathcal{X})$ is determined by the number of nonzero singular tubes of in $\mathcal{S}$ obtained from the T-SVD of $\mathcal{X}$. Specifically, $rank_t(\mathcal{X}) = \#\{l, \mathcal{S}(l,l,:) \neq \mathbf{0}\}$.*

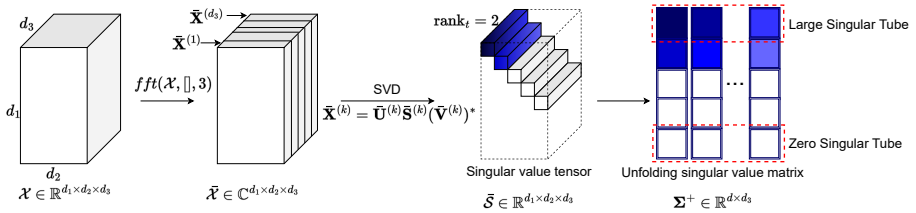


Figure 3: An illustration of the unfolding singular value matrix $\mathbf{\Sigma}^+$ of a low-tubal-rank tensor $\mathcal{X}$ ($\text{rank}_t(\mathcal{X}) = 2$). Note that entries of each row of $\mathbf{\Sigma}^+$ come from the same singular tube in the tensor $\bar{\mathcal{S}}$, where $\mathcal{S}$ is the f-diagonal tensor in the T-SVD of $\mathcal{X} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$ and $\bar{\mathcal{S}} = fft(\mathcal{S}, [], 3)$. Therefore, $\mathbf{\Sigma}^+$ is joint row sparse, i.e., the singular value tensor $\bar{\mathcal{S}}$ equips with the tube sparse structure.

Definition 6. *(**Tensor Nuclear Norm** [7]) Given $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ and $d = \min(d_1, d_2)$, the tensor nuclear norm (TNN) is given by*

$$\|\mathcal{X}\|_* = \frac{1}{d_3} \sum_{k=1}^{d_3} \|\bar{\mathbf{X}}^{(k)}\|_* = \frac{1}{d_3} \sum_{k=1}^{d_3} \sum_{l=1}^{d} \sigma_l \left(\bar{\mathbf{X}}^{(k)}\right). \tag{7}$$

3. Proposed Method

3.1. Motivation

As defined in Definition 6, a key insight is that TNN can be induced by the sparsity measure of the tensor singular values. To clarify this, we first introduce the definition of the unfolding singular value matrix.

Definition 7. *(**Unfolding Singular Value Matrix**) Given a tensor $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, its unfolding singular value matrix $\mathbf{\Sigma}^+ \in \mathbb{R}^{d \times d_3}$ is defined as*

$$\mathbf{\Sigma}^+ = \begin{bmatrix} \sigma_1^{(1)} & \sigma_1^{(2)} & \cdots & \sigma_1^{(d_3)} \\ \sigma_2^{(1)} & \sigma_2^{(2)} & \cdots & \sigma_2^{(d_3)} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_d^{(1)} & \sigma_d^{(2)} & \cdots & \sigma_d^{(d_3)} \end{bmatrix},$$

where $\sigma_l^{(k)} = \sigma_l \left(\bar{\mathbf{X}}^{(k)}\right)$, $d = \min(d_1, d_2)$, and $\bar{\mathcal{X}} = fft(\mathcal{X}, [], 3)$.

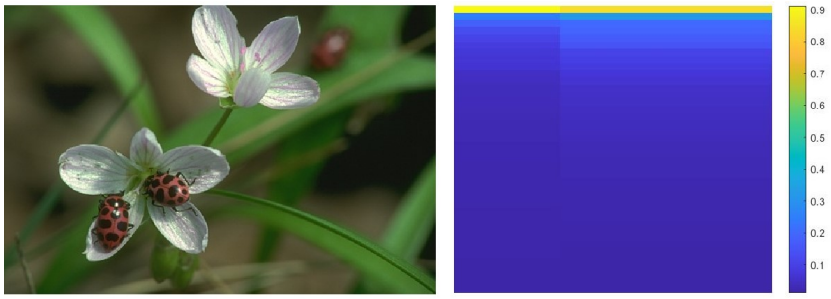


Figure 4: The unfolding singular value matrix of a color image is joint row sparse. (a) A color image can be denoted as $\mathcal{X} \in \mathbb{R}^{321\times481\times3}$; (b) unfolding singular value matrix $\mathbf{\Sigma}^+ \in \mathbb{R}^{321\times3}$ of $\mathcal{X}$ (For better understanding, we only visualize the first 40 rows of $\mathbf{\Sigma}^+$ since other rows are all zero vectors).

Based on Definition 7, TNN can be reformulated as a scaled ℓ_1 norm of the unfolding singular value matrix $\mathbf{\Sigma}^+$, i.e., $\|\mathcal{X}\|_* = \frac{1}{d_3}\|\mathbf{\Sigma}^+\|_1$. As illustrated in Fig. 3, the l-th row of $\mathbf{\Sigma}^+$, $[\sigma_l^{(1)}, \ldots, \sigma_l^{(d_3)}]$, consists of the singular values associated with the l-th singular tube of $\bar{\mathcal{S}}$. Therefore, a singular tube is zero if and only if all entries in the corresponding row of $\mathbf{\Sigma}^+$ are zero. According to the definition of tubal rank, a low-tubal-rank tensor satisfying $\text{rank}_t(\mathcal{X}) \ll d = \min(d_1, d_2)$ has only a few nonzero rows in its unfolding singular value matrix $\mathbf{\Sigma}^+$. In other words, the row sparsity of $\mathbf{\Sigma}^+$ provides an equivalent representation of the low-tubal-rank property of $\mathcal{X}$. As illustrated in Fig. 4, the unfolding singular value matrix $\mathbf{\Sigma}^+ \in \mathbb{R}^{321\times3}$ of a color image exhibits a clear joint row-sparse pattern, with only a few rows being nonzero. Although TNN has achieved promising performance in low-rank tensor recovery, it characterizes tensor rank through the sum of singular values and cannot fully capture the inherent tube-sparse property of the singular value tensor.

Another crucial observation is that different singular tubes often have distinct physical meanings in real applications. As reported in [8, 27], dominant singular tubes usually capture important components of the underlying low-rank tensor, such as color characteristics in images [8]. Therefore, they should be penalized less to avoid excessive information loss. When the large singular tubes are penalized less and the small ones are penalized more, it further encourages the tube sparse structure of singular

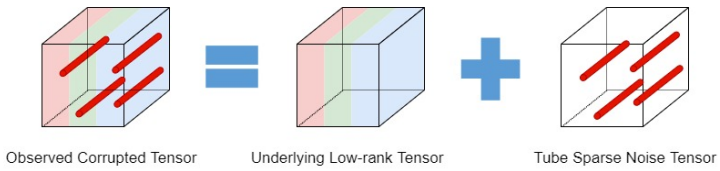


Figure 5: Illustration of the TRPCA model against tube sparse noise.

value tensor.

Moreover, as pointed out in [28, 29], sparse noise often exhibits a tube sparse structure in many real-world tensor data, as illustrated in Fig. 5, including color images, grayscale videos, etc. For example, correlated color channels may suffer from simultaneous corruption at identical spatial locations due to their inherent structural dependencies. However, the classical tensor ℓ_1 norm penalizes each entry independently, without accounting for the correlation among entries within the same tube. As a result, it fails to preserve the tube sparse structure of the noise tensor, thereby failing to fully exploit its intrinsic structural information.

To address these limitations, we design two tube-preserving regularizers, T2NN and T2L1, which respectively preserve the tube sparse structure of the singular value tensor and the noise tensor. Building on these regularizers, we develop a novel TRPCA model, referred to as Double Tube-preserving Regularization (DTR), to more effectively characterize the underlying data structure.

3.2. TRPCA with Double Tube-preserving Regularization

As mentioned above, we should preserve the tube sparse structure of the singular value tensor. Moreover, singular tubes with larger magnitudes require milder penalization to avoid excessive loss of important structures in tensor data. Accordingly, we define the tube-preserving tensor nuclear norm (T2NN).

Definition 8. ***(Tube-preserving Tensor Nuclear Norm, T2NN)*** *For $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, we define its tube-preserving tensor nuclear norm as*

$$\|\mathcal{X}\|_{T,*} := \frac{1}{d_3} \sum_{l=1}^{d} f_{\mathcal{L}}\left(\left\|\vec{\boldsymbol{\sigma}}_l(\bar{\mathcal{X}})\right\|_2\right) \sum_{k=1}^{d_3} \sigma_l\left(\bar{\mathbf{X}}^{(k)}\right), \tag{8}$$

where $d = \min(d_1, d_2)$, $\vec{\boldsymbol{\sigma}}_l(\bar{\mathcal{X}}) = \left[\sigma_l\left(\bar{\mathbf{X}}^{(1)}\right), \cdots, \sigma_l\left(\bar{\mathbf{X}}^{(d_3)}\right)\right] \in \mathbb{R}^{d_3}$ *is the* l*-th singular tube of* $\mathcal{X}$ *for* $l \in [d]$*, and the structure function* $f_{\mathcal{L}}(\cdot)$ *is given by* $f_{\mathcal{L}}\left(\left\|\vec{\boldsymbol{\sigma}}_l(\bar{\mathcal{X}})\right\|_2\right) = 1/\left(1 + \left\|\vec{\boldsymbol{\sigma}}_l(\bar{\mathcal{X}})\right\|_2 / \tau_{\mathcal{L}}\right)$*, where* $\tau_{\mathcal{L}}$ *is a positive parameter.*

Remark 1. *T2NN preserves the tube sparse structure of the singular value tensor in two aspects. First, by assigning equivalent* $f_{\mathcal{L}}(\cdot)$ *on the singular values of the same singular tube* $\vec{\boldsymbol{\sigma}}_l(\bar{\mathcal{X}})$*, nonzero singular values are holistically regularized on tubes rather than treated independently, which better preserves the tube sparse structure of the singular value tensor compared with TNN. Second, since larger singular tubes carry more prominent information [8],* $f_{\mathcal{L}}(\cdot)$ *imposes smaller penalization on larger singular tubes to preserve the prominent information carried by tensor data.*

Additionally, the error modeling term for the noise tensor can be flexible in our framework. When the noise tensor is sparse (see Fig. 1), we can choose classical tensor ℓ_1 norm. However, as discussed in Section 3.1, the noise tensor may also exhibit a tube sparse structure, which the classical ℓ_1 norm fails to preserve. Similarly, we introduce the Tube-preserving Tensor ℓ_1 norm to exploit the tube sparse structure of the noise tensor.

Definition 9. *(**Tube-preserving Tensor** ℓ_1 **Norm, T2L1**) For* $\mathcal{E} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$*, we define its tube-preserving tensor* ℓ_1 *norm as*

$$\|\mathcal{E}\|_{T,1} := \sum_{i=1}^{d_1} \sum_{j=1}^{d_2} f_{\mathcal{E}}\left(\|\mathcal{E}(i,j,:)\|_2\right) \sum_{k=1}^{d_3} |\mathcal{E}_{ijk}|, \tag{9}$$

where $\mathcal{E}(i,j,:) = \left[\mathcal{E}_{ij1}, \cdots, \mathcal{E}_{ijd_3}\right] \in \mathbb{R}^{d_3}$ *is the* (i,j)*-th noise tube for* $i \in [d_1], j \in [d_2]$*, and structure function* $f_{\mathcal{E}}(\cdot)$ *is given by* $f_{\mathcal{E}}\left(\|\mathcal{E}(i,j,:)\|_2\right) = 1/\left(1 + \|\mathcal{E}(i,j,:)\|_2 / \tau_{\mathcal{E}}\right)$*, where* $\tau_{\mathcal{E}}$ *is a positive parameter.*

Remark 2. *Most previous works employ the tensor* ℓ_1 *norm to model sparse noise. However, when the noise is tube-wise, such methods are unable to preserve the intrinsic tube-sparse structure exhibited by the noise tensor. By comparison, T2L1 regularizes noise tubes holistically by assigning a common penalty to all entries within the same noise tube, thereby better preserving the tube-sparse structure of the noise tensor.*

Objective function. Equipped with the newly defined T2NN and T2L1, we propose a novel TRPCA model based on double tube-preserving regularization, termed DTR, with the following objective function:

$$\min_{\mathcal{L},\mathcal{E}\in\mathbb{R}^{d_1\times d_2\times d_3}} \|\mathcal{L}\|_{T,*} + \lambda\|\mathcal{E}\|_{T,1} \ \text{ s.t. } \ \mathcal{X} = \mathcal{L} + \mathcal{E}. \tag{10}$$

3.3. Optimization for DTR

The corresponding Lagrangian function for DTR (10) is expressed as

$$\begin{aligned} L(\mathcal{L},\mathcal{E},\mathcal{N},\mu) &= \|\mathcal{L}\|_{T,*} + \lambda\|\mathcal{E}\|_{T,1} \\ &+ \frac{\mu}{2}\left\|\mathcal{L}+\mathcal{E}-\mathcal{X}+\mathcal{N}/\mu\right\|_F^2 - \frac{\mu}{2}\left\|\mathcal{N}/\mu\right\|_F^2. \end{aligned} \tag{11}$$

Here, $\mu > 0$ is a positive scalar parameter, and $\mathcal{N}$ denotes the Lagrange multiplier. We minimize Eq. (11) iteratively by fixing each variable and adjusting the others. Based on the ADMM framework [26], the subproblems are updated as follows:

To facilitate the derivation of the update steps, we define the adaptive weights of the structure functions as

$$w_l = f_{\mathcal{L}}\left(\left\|\vec{\sigma}_l(\bar{\mathcal{L}})\right\|_2\right), \quad W_{ij} = f_{\mathcal{E}}\left(\|\mathcal{E}(i,j,:)\|_2\right),$$

where $l \in [d]$, $d = \min(d_1, d_2)$ for w_l, and $i \in [d_1]$, $j \in [d_2]$ for W_{ij}. Here, w_l denotes the adaptive weight for the l-th singular value tube of $\bar{\mathcal{L}}$, and W_{ij} represents the adaptive weight for the (i, j)-th tube of $\mathcal{E}$. These values are assembled into a weight vector $\mathbf{w} = [w_1, \cdots, w_d] \in \mathbb{R}^d$ and a weight matrix $\mathbf{W} \in \mathbb{R}^{d_1\times d_2}$.

Step 1: Update $\mathcal{L}$ with the remaining variables fixed:

$$\min_{\mathcal{L}\in\mathbb{R}^{d_1\times d_2\times d_3}} \frac{1}{2}\left\|\mathcal{L}-(\mathcal{X}-\mathcal{E}_t-\mathcal{N}_t/\mu_t)\right\|_F^2 + \frac{1}{\mu_t}\|\mathcal{L}\|_{T,*}. \tag{12}$$

For problem (12), the following theorem gives the closed-form solution.

Given a tensor $\mathcal{M} \in \mathbb{R}^{d_1\times d_2\times d_3}$ and the vector $\mathbf{w} = [w_1, \cdots, w_d] \in \mathbb{R}^d$, with $d = min(d_1, d_2)$, and $\mathcal{M} = \mathcal{U} * \mathcal{S} * \mathcal{V}^*$ denote its T-SVD. We give the definition of the proximal operator of T2NN for each $\tau \geq 0$ as

$$\mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}) = \mathcal{U} * ifft\left(\mathcal{P}_{\tau\mathbf{w}}(\bar{\mathcal{S}}), [], 3\right) * \mathcal{V}^*. \tag{13}$$

Here, $ifft(\cdot,[],3)$ denotes the inverse fast Fourier transform along the third dimension. For each $k \in [d_3]$, the matrix SVD of $\bar{\mathbf{M}}^{(k)}$ yields the singular value matrix $\bar{\mathbf{S}}^{(k)}$, which is then transformed by $\mathbf{P}_{\tau\mathbf{w}}$ to form the k-th frontal slice $\mathbf{P}_{\tau\mathbf{w}}(\bar{\mathbf{S}}^{(k)})$ of $\mathcal{P}_{\tau\mathbf{w}}(\bar{\mathcal{S}})$. Here $\mathbf{P}_{\tau\mathbf{w}}\left(\bar{\mathbf{S}}^{(k)}\right)$ is diagonal and has the same dimensions as $\bar{\mathbf{S}}^{(k)}$ and $\left(\mathbf{P}_{\tau\mathbf{w}}\left(\bar{\mathbf{S}}^{(k)}\right)\right)_{ll} = (\bar{\mathbf{S}}^{(k)}_{ll} - \tau \cdot w_l)_+$, where $(z)_+ = z$ if $z > 0$ and $(z)_+ = 0$ if $z \leq 0$.

Theorem 1. *Given $\tau \geq 0$ and a weight vector $\mathbf{w} \in \mathbb{R}^d$, if $\mathbf{w}$ satisfies $0 \leq w_1 \leq \cdots \leq w_d$, the T2NN proximal operator (13) of the tensor $\mathcal{M} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$ obeys*

$$\mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}) = \underset{\mathcal{L} \in \mathbb{R}^{d_1 \times d_2 \times d_3}}{\arg\min} \frac{1}{2}\|\mathcal{L} - \mathcal{M}\|_F^2 + \tau\|\mathcal{L}\|_{T,*}, \tag{14}$$

where $d = min(d_1, d_2)$.

Proof. The proof is seen in the supplementary material. □

It follows from Theorem 1 that the subproblem for $\mathcal{L}$ in Eq. (12) is $\mathcal{L}_{t+1} = \mathbf{prox}_{\|\cdot\|_{T,*}}(\mathcal{M}_t)$, where $\mathcal{M}_t = \mathcal{X} - \mathcal{E}_t - \mathcal{N}_t/\mu_t$ and $\tau = 1/\mu_t$.

Step 2: Update $\mathcal{E}$ with the remaining variables fixed:

$$\min_{\mathcal{E} \in \mathbb{R}^{d_1 \times d_2 \times d_3}} \frac{1}{2}\|\mathcal{E} - (\mathcal{X} - \mathcal{L}_{t+1} - \mathcal{N}_t/\mu_t)\|_F^2 + \frac{\lambda}{\mu_t}\|\mathcal{E}\|_{T,1}. \tag{15}$$

We now define a tensor soft-thresholding operator $\mathcal{Q}$ such that $(\mathcal{Q}(\mathcal{K}, \mathbf{W}))_{ijk} = \text{sign}(\mathcal{K}_{ijk})(|\mathcal{K}_{ijk}| - W_{ij})_+$. Then Eq. (15) is solved by

$$\mathcal{E}_{t+1} = \mathcal{Q}\left(\mathcal{K}_t, \frac{\lambda}{\mu_t}\mathbf{W}_t\right), \ \mathcal{K}_t = \mathcal{X} - \mathcal{L}_{t+1} - \mathcal{N}_t/\mu_t. \tag{16}$$

Step 3: We first update the weight vector $\mathbf{w}$ for singular tubes of $\bar{\mathcal{L}}$ by

$$w_l = \frac{1}{1 + \left\|\vec{\sigma}_l(\bar{\mathcal{L}}_{t+1})\right\|_2 / \tau_{\mathcal{L}}}, \ l \in [d], \tag{17}$$

where $[d] = \{1, 2, \ldots, d\}$. We then update the weight matrix $\mathbf{W}$ of $\mathcal{E}$ by

$$W_{ij} = \frac{1}{1 + \|\mathcal{E}_{t+1}(i, j, :)\|_2 / \tau_{\mathcal{E}}}, \ i \in [d_1], \ j \in [d_2], \tag{18}$$

where $\tau_{\mathcal{L}}$ and $\tau_{\mathcal{E}}$ are positive hyperparameters whose settings are detailed in the parameter analysis below. Note that $\mathbf{w}$ and $\mathbf{W}$ are iteratively updated during optimization,

Algorithm 1 TRPCA via Double Tube-preserving Regularization (DTR)

Input: Observation $\mathcal{X} \in \mathbb{R}^{d_1 \times d_2 \times d_3}$, and the parameter λ.

Initialize: $\mathcal{L}_0 = \mathbf{0}$, $\mathcal{E}_0 = \mathbf{0}$, $\mathcal{N}_0 = \mathbf{0}$, $\mathbf{w}_0 = \mathbf{1}_d$, $\mathbf{W}_0 = \mathbf{1}_{d_1 \times d_2}$, $\mu_0 = 10^{-2}$, $\rho = 1.1$, $\epsilon = 10^{-6}$ and $t = 0$.

while not converged **do**

1: Update $\mathcal{L}_{t+1}$ by Eq. (12).

2: Update $\mathcal{E}_{t+1}$ by Eq. (15).

3: Update $\mathbf{w}_{t+1}$ and $\mathbf{W}_{t+1}$ by Eqs. (17) and (18), respectively.

4: Update $\mathcal{N}_{t+1} = \mathcal{N}_t + \mu_t(\mathcal{L}_{t+1} + \mathcal{E}_{t+1} - \mathcal{X})$.

5: Update μ_{t+1} by $\mu_{t+1} = \rho\mu_t$.

6: Check the convergence conditions:

$$\|\mathcal{L}_{t+1} - \mathcal{L}_t\|_\infty < \epsilon, \|\mathcal{E}_{t+1} - \mathcal{E}_t\|_\infty < \epsilon, \|\mathcal{X} - \mathcal{L}_{t+1} - \mathcal{E}_{t+1}\|_\infty < \epsilon.$$

end while

Output: $\mathcal{L} = \mathcal{L}_{t+1}$, $\mathcal{E} = \mathcal{E}_{t+1}$.

which enables the proposed algorithm to automatically assign smaller penalties to significant singular value tubes and important sparse tubes, thereby better preserving the underlying low-rank and sparse structures.

Step 4: We can update $\mathcal{N}$ and μ by

$$\mathcal{N}_{t+1} = \mathcal{N}_t + \mu_t(\mathcal{L}_{t+1} + \mathcal{E}_{t+1} - \mathcal{X}),\ \mu_{t+1} = \rho\mu_t,\ \rho = 1.1.$$

Algorithm 1 provides the pseudo-code of DTR.

Computational Complexity Analysis: At each iteration, the computational cost mainly lies on the update of $\mathcal{L}_{t+1}$. For simplicity, let $d_{(1)} = \max\{d_1, d_2\}$ and $d_{(2)} = \min\{d_1, d_2\}$. Since update of $\mathcal{L}_{t+1}$ needs to compute Fourier transform and the matrix SVD, its computational cost per iteration is $O(d_1 d_2 d_3 \log d_3 + d_{(1)} d_{(2)}^2 d_3)$. Note that such

cost is the same order to the baseline TNN minimization algorithm in [7].

Parameter Analysis: How to automatically select proper hyper-parameters $\tau_{\mathcal{L}}$ and $\tau_{\mathcal{E}}$ in the low-rank and sparse regularization terms remains a challenging problem. The optimal parameters for our proposed regularization terms vary across different tensor datasets. Specifically, $\tau_{\mathcal{L}}$ controls the function values associated with different singular tubes. When $\tau_{\mathcal{L}}$ is too large, close to ∞, T2NN reduces to the original TNN and the tube sparse structure is damaged. When $\tau_{\mathcal{L}}$ is too small, all the function values would be close to 0, resulting in the collapse of the low-rank structure. Hence, $\tau_{\mathcal{L}}$ may be vastly different for various tensor data. This makes manually searching the optimal parameters extremely difficult. To address this issue, we set $\tau_{\mathcal{L}} = \gamma_{\mathcal{L}} * \text{mean}(\left\|\vec{\boldsymbol{\sigma}}_1(\bar{\mathcal{L}})\right\|_2, \cdots, \left\|\vec{\boldsymbol{\sigma}}_d(\bar{\mathcal{L}})\right\|_2)$, where $\gamma_{\mathcal{L}}$ is a positive constant. By this way, $\tau_{\mathcal{L}}$ is set to a proper value and the strategy of estimation by the mean value can be effective. Our strategy is user-friendly and originated from robust learning [30, 31]. Similarly, we set $\tau_{\mathcal{E}} = \gamma_{\mathcal{E}} * \text{mean}(\|\mathcal{E}(1,1,:)\|_2, \cdots, \|\mathcal{E}(d_1,d_2,:)\|_2)$, where $\gamma_{\mathcal{E}}$ is a positive constant. The specific settings of the $\gamma_{\mathcal{L}}$ and $\gamma_{\mathcal{S}}$ are described in the experiments.

3.4. Convergence Analysis

This subsection devotes to establishing the convergence analysis for Algorithm 1. Note that the convergence analysis of previous works such as [8] mainly focuses on the models which employ the fixed and predefined weight variables. Besides, most of them contain only single weighted regularization term. Unlike these works, both low-rank and sparse regularization terms are weighted functions while the double weight variables are dynamic and learned in the iterations. Hence, the previous results and proof techniques of the convergence analysis cannot be directly applied to our model. This makes the convergence analysis more challenging and difficult.

Lemma 1. *The sequences* $\{\mathcal{N}_t\}_{t=1}^{\infty}$, $\{\mathbf{w}_t\}_{t=1}^{\infty}$, $\{\mathbf{W}_t\}_{t=1}^{\infty}$ *generated by Algorithm 1 are bounded.*

Proof. The proof is seen in the supplementary material. □

Lemma 2. *The sequences* $\{\mathcal{L}_t\}_{t=1}^{\infty}$, $\{\mathcal{E}_t\}_{t=1}^{\infty}$ *generated by Algorithm 1 are bounded.*

Proof. The proof is seen in the supplementary material. □

With the aforementioned results, we are now ready to prove that the proposed algorithm is convergent.

Theorem 2. *Algorithm 1 generates the sequences* $\{\mathcal{L}_t\}_{t=1}^{\infty}$, $\{\mathcal{E}_t\}_{t=1}^{\infty}$ *satisfying*
(1) $\lim_{t\to\infty} \|\mathcal{L}_t + \mathcal{E}_t - \mathcal{X}\|_F = 0$.
(2) $\lim_{t\to\infty} \|\mathcal{L}_{t+1} - \mathcal{L}_t\|_F = 0$.
(3) $\lim_{t\to\infty} \|\mathcal{E}_{t+1} - \mathcal{E}_t\|_F = 0$.

Proof. The proof is provided in the supplementary material. □

3.5. Extension of DTR with Smoothness Prior

In this part, we extent DTR with smoothness prior to enhance its recovery capability. To begin with, we first introduce the t-CTV [21], i.e., a unified regularizer of joint low-rankness and smoothness.

Definition 10. ***(Gradient tensor**[21]) Givn* $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$, *we denote the gradient along the m-th mode by*

$$\mathcal{G}_m = \nabla_m(\mathcal{X}) = \mathcal{X} \times_m \mathbf{D}_{d_m},\ m = 1, 2, 3, \tag{19}$$

where $\times_m$ *denotes mode-m product [4] and* $\mathbf{D}_{d_m}$ *is a row circulant matrix of* $(-1, 1, 0, \cdots, 0)$.

Accordingly, the t-CTV regularizer can be formulated as

$$\|\mathcal{X}\|_{\text{t-CTV}} := \frac{1}{|\Gamma|}\sum_{m\in\Gamma} \|\mathcal{G}_m\|_*, \tag{20}$$

where Γ collects the directions along which $\mathcal{X}$ exhibits smoothness and $|\Gamma|$ denotes the cardinality of Γ. Note that Γ is a priori set given from the concrete visual tensor data. For example, color images typically exhibit smoothness in the spatial dimensions, corresponding to $\Gamma = 1, 2$. For grayscale videos, Γ can be set as $\{1, 2, 3\}$.

This shows that t-CTV is also based on the TNN framework and imposes the TNN regularization to the gradient tensors. Similarly, we can enhance t-CTV by considering the tube sparse structure of singular value tensor of $\mathcal{G}_m$.

Definition 11. ***(Tube-preserving Gradient Tensor Nuclear Norm, TGTNN)*** *For* $\mathcal{X} \in \mathbb{R}^{d_1\times d_2\times d_3}$, *we define the Tube-preserving Gradient Tensor Nuclear Norm as*

$$\|\mathcal{X}\|_{TG,*} = \frac{1}{|\Gamma|}\sum_{m\in\Gamma} \|\mathcal{G}_m\|_{T,*}, \tag{21}$$

where Γ *collects the directions along which* $\mathcal{X}$ *exhibits smoothness and* $|\Gamma|$ *denotes the cardinality of* Γ.

Equipped with TGTNN, we can extend DTR to a joint learning model of low-rankness and smoothness:

$$\min_{\mathcal{L},\mathcal{E}\in\mathbb{R}^{d_1\times d_2\times d_3}} \|\mathcal{L}\|_{TG,*} + \lambda\|\mathcal{E}\|_{T,1} \ \text{ s.t. } \ \mathcal{X} = \mathcal{L} + \mathcal{E}. \tag{22}$$

We call the model (22) as Gradient DTR (GDTR). The optimization of GDTR is similar to DTR and the algorithm can be viewed as a combination of DTR and t-CTV, which can be seen in the supplementary material.

4. Experiments

We apply DTR and GDTR for various real-world applications. **Due to space constraints, further experimental results are reported in the supplementary material.**

4.1. Application to Color Image Recovery

We substantiate the superiority of DTR and GDTR on color image recovery. The Berkeley Segmentation Dataset and Benchmark 500 (BSDS500) [32] is employed as a benchmark to evaluate the performance of DTR and GDTR. It consists of 500 natural color images, with 200, 100, and 200 images allocated to the training, validation, and test sets, respectively. Following previous image restoration studies, we use the test subset for evaluation. Following [7], we corrupt each color image by randomly replacing 10% of its pixels with arbitrary values drawn from [0, 255]. The corruptions are on the whole tubes and all 3 channels are corrupted at the same positions.

Comparison methods: We evaluate DTR and GDTR with several state-of-the-art approaches, including TRPCA [7], ETRPCA [8], BTRTF [33], t-SWP [34], SRPCA [35], BTPCA [36], and t-CTV [21]. All compared methods are implemented using the parameter settings recommended in their original papers. Specifically, the regularization parameters of TRPCA and ETRPCA are set as $1/\sqrt{d_3\max(d_1,d_2)}$, while that of t-CTV is set as $1/\sqrt{d_1d_2d_3/\min(d_1,d_2)}$. The Schatten-$p$ parameter of t-SWP is set to $p = 0.9$, and the initial multi-rank and outlier variance of BTRTF are set to 150 and

Index	1	2	3	4	5	6
TRPCA	31.05	33.95	28.25	25.49	27.12	29.54
ETRPCA	31.73	34.62	29.08	26.30	27.74	30.30
BTRTF	32.94	33.34	31.50	28.99	28.91	32.88
t-SWP	34.76	36.93	31.64	29.56	29.74	33.79
SRPCA	31.12	33.92	28.37	25.80	27.48	29.71
BTPCA	32.67	35.56	30.06	26.76	29.21	31.99
DTR	34.31	37.13	32.19	29.42	30.16	33.40
t-CTV	34.54	37.76	33.04	30.29	29.55	33.13
GDTR	**37.44**	**40.32**	**35.92**	**33.60**	**32.22**	**36.61**

Table 2: The PSNR values of the recovered results by various methods on 6 example images which are shown in Fig. 6. Best results are marked bold.

10^{-7}, respectively. For DTR and GDTR, $\lambda = 1/\sqrt{d_3 \max(d_1, d_2)}$ is adopted according to the theoretical analysis, with $\gamma_{\mathcal{L}} = 3$ and $\gamma_{\mathcal{E}} = 10^3$. We adopt the Peak Signal-to-Noise Ratio (PSNR) to quantify the quality of recovery images [37].

Results: Fig. 6 and Tab. 2 present several examples of the recovered results obtained from the BSD dataset, together with their corresponding PSNR values and computational times. To further provide a comprehensive evaluation, we apply all competing methods to the entire BSD dataset, with the results reported in Tab. 3. DTR achieves better recovery performance than TRPCA, which relies solely on the low-rankness prior. This improvement can be attributed to the tube-preserving regularization, which exploits the tube-sparse structures of both the singular value tensor and the corruption tensor. Furthermore, GDTR consistently achieves superior recovery performance compared with the competing methods. Although t-CTV has demonstrated the advantages of jointly exploiting low-rankness and smoothness priors, GDTR consistently yields better recovery results on the color image data, highlighting the contribution of the smoothness prior to the proposed method.

Furthermore, we evaluate the recovery performance under different corruption ratios ranging from 10% to 50%. The quantitative results for BSDval are reported in Tab. 4. As the corruption ratio increases, the PSNR values generally decrease for all

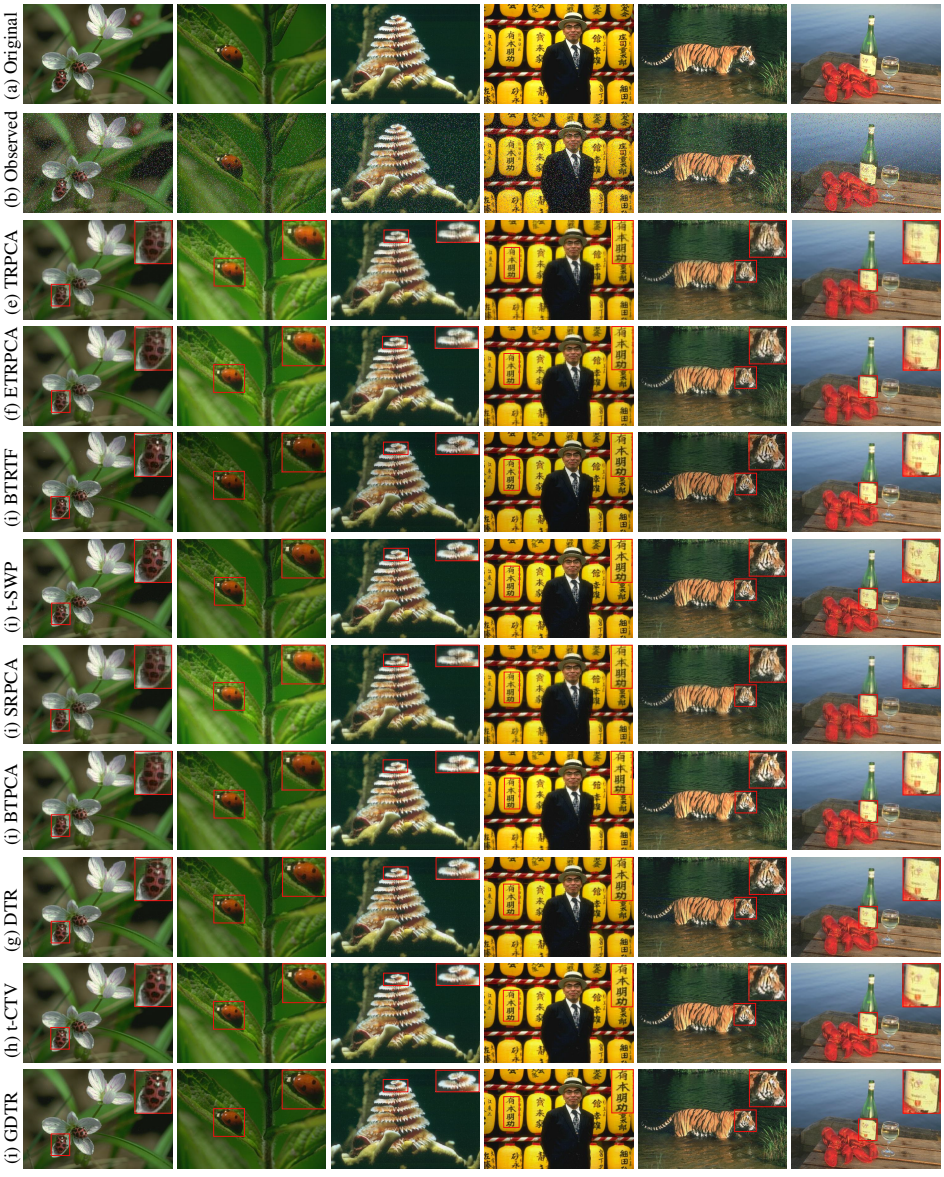


Figure 6: Qualitative comparison of image recovery results for six representative BSD samples. (a) Original image; (b) observed image; (c)-(i) recovered images by various methods. The quantitative results in terms of PSNR are reported in Tab. 2. For better visual comparison, please zoom in to 2× in the color PDF.

Methods	BSDtrain (200)	BSDval (100)	BSDtest (200)	BSD (500)	Avg. Time (s)
TRPCA	29.11	28.72	28.63	28.84	**5.13**
ETRPCA	29.50	29.06	29.02	29.22	6.02
BTRTF	29.17	28.33	28.92	28.90	17.85
t-SWP	30.96	30.32	30.53	30.66	12.26
SRPCA	29.24	28.86	28.78	28.98	10.94
BTPCA	30.28	29.82	29.80	30.00	7.50
DTR	31.14	30.53	30.70	30.84	6.68
t-CTV	31.40	30.79	30.85	31.06	14.93
GDTR	**33.36**	**32.59**	**32.94**	**33.03**	18.26

Table 3: Average PSNR values and computation time (in seconds) obtained on the BSD dataset and its subsets under 10% corruption. The best results are highlighted in bold.

methods. Nevertheless, GDTR consistently achieves the highest PSNR under different corruption ratios, and the improvement becomes more pronounced under heavier corruption. These results further verifies the robustness of GDTR against varying levels of tube-sparse corruption.

4.2. Application to Hyperspectral Image Restoration

To further evaluate the accuracy and efficiency of DTR and GDTR, we apply them to hyperspectral image (HSI) restoration. We conduct experiments on a more large-scale tensor dataset, the Pavia university (PaviaU) dataset, the size of which is $610 \times 340 \times 100$. We randomly select 10% of the pixels and replace them with samples from $[0, 255]$, and the same pixel locations are corrupted across all 100 spectral bands. The competing methods are the same as those used in the color image recovery. The comparison of restoration performance and computation time of various methods for the HSI restoration on the PaviaU dataset is shown in Fig. 7. Considering both the performance and time cost, DTR outperforms other competing methods. Besides, although the time cost of GDTR is more than others which is still acceptable, it achieves the best performance.

Methods	10%	20%	30%	40%	50%	Avg. Time (s)
TRPCA	28.73	26.07	23.27	20.66	18.27	4.56
ETRPCA	29.08	26.44	23.51	20.70	18.12	**3.69**
BTRTF	28.34	24.56	21.07	18.12	15.74	17.68
t-SWP	30.32	26.66	23.07	19.76	17.02	8.45
SRPCA	28.87	26.17	23.33	20.69	18.29	7.61
BTPCA	29.82	26.98	23.88	20.85	17.61	5.93
DTR	31.14	28.10	24.46	20.68	19.53	8.23
t-CTV	32.29	28.63	24.89	21.66	19.03	22.01
GDTR	**33.36**	**29.25**	**25.54**	**25.30**	**23.62**	27.19

Table 4: Average PSNR values and computation time (in seconds) obtained on the BSDval(100) subset with corrupted ratios from 10% to 50%. The best results are highlighted in bold.

4.3. Application to Video Denoising

This experiment devotes to evaluate the performance of DTR and GDTR for video denoising, which aims to recover the clean grayscale video against the tube sparse noise. We conduct experiments on the YUV [38] video sequences and take 4 videos from the sequences including Akiyo, Claire, Container and News.The evaluation is conducted using the initial 100 frames from each video sequence and each grayscale video can be viewed as a $144 \times 176 \times 100$ tensor. We randomly select 10% of the pixels and replace them with samples from $[0, 255]$, and all the 100 frames of the videos are corrupted at the same positions. This setting poses greater challenges than independent corruptions occurring at different locations across frames.

Tab. 5 presents the quantitative results and computation time in average of each method over four video sequences of the YUV dataset. To further illustrate the denoising performance of DTR and GDTR, an example frame from three video sequences is presented in Fig. 8. The above observations can be summarized as follows:

- Firstly, DTR outperforms comparison methods in light of the PSNR value and recovery results on video denoising. The reason is that it simultaneously considers the tube sparse structure of the singular value tensor and noise tensor. Exploiting this prior information, our method can more accurately separate the underlying

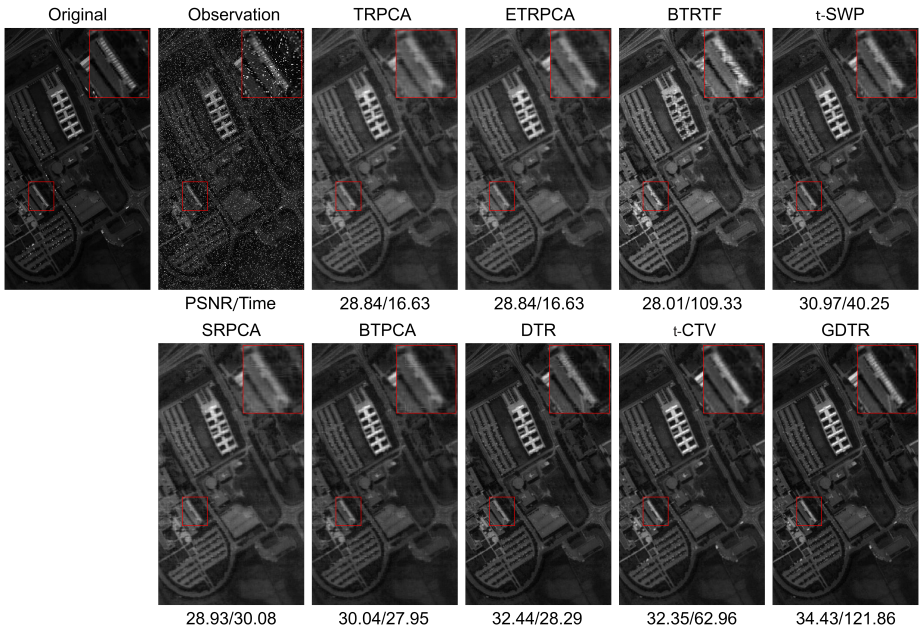


Figure 7: Comparison of denoising performance and computation time (in seconds) of various methods for the HSI restoration on the PaviaU dataset. For better visual comparison, please zoom in to 2× in the color PDF.

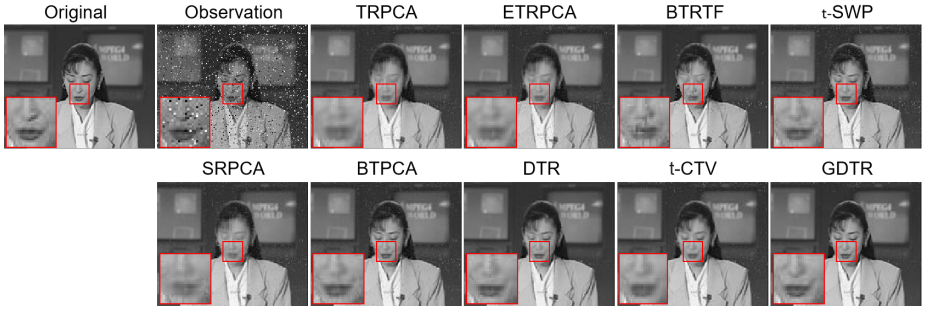
low-rank component and sparse noise tensor.

- Secondly, due to the increasing of 3-rd dimension and the tube-wise noise, the task can be more challenging. However, DTR and GDTR can still achieve superior performance than other competing methods.

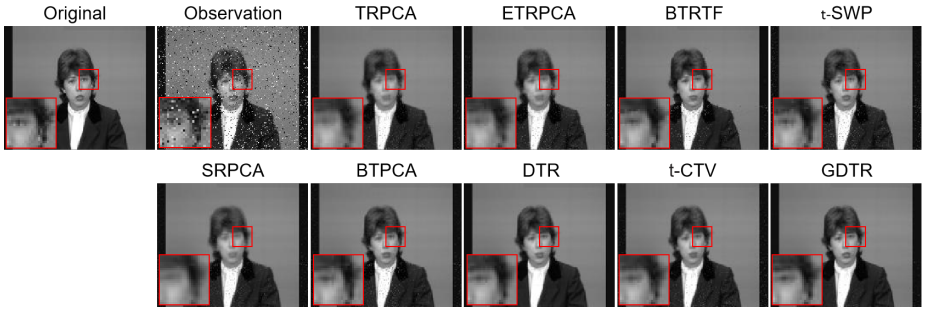
4.4. Parameter Discussion and Ablation Study

To further investigate the effects of different parameters and regularization terms, we perform parameter analysis and ablation studies.

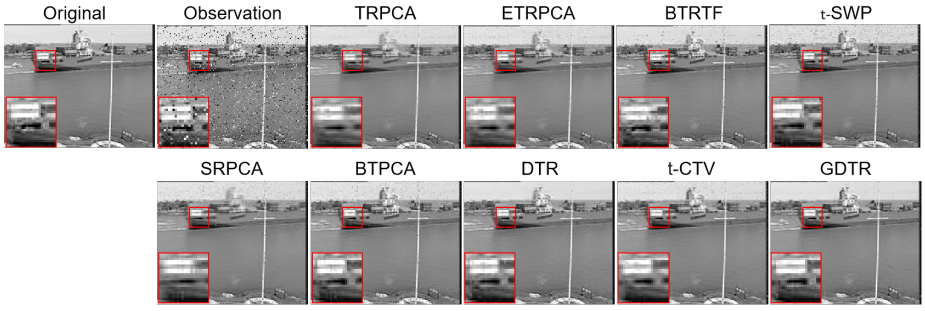
Parameter Analysis: We investigate how the choices of $\gamma_{\mathcal{L}}$ and $\gamma_{\mathcal{E}}$ affect denoising performance, where they control the structure functions of the low-rank and sparse regularization terms, respectively. Fig. 9 illustrates the parameter variations in color image recovery under different settings of $\gamma_{\mathcal{L}}$ and $\gamma_{\mathcal{E}}$.



(a) Akiyo



(b) Claire



(c) Container

Figure 8: Comparison of denoising performance on 3 video sequences form the YUV dataset. (a) Akiyo; (b) Claire; (c) Container. For better visual comparison, please zoom in to 2× in the color PDF.

Methods	Akiyo	Claire	Container	News	Avg. Time (s)
TRPCA	30.24	30.12	28.49	26.01	**15.19**
ETRPCA	30.38	30.30	28.77	26.33	15.30
BTRTF	30.76	29.24	28.23	25.02	46.78
t-SWP	31.17	30.11	28.93	26.56	38.32
SRPCA	30.77	30.41	28.10	26.25	23.07
BTPCA	31.82	31.56	30.02	27.74	34.53
DTR	33.57	35.12	31.85	29.39	26.64
t-CTV	32.24	32.32	31.07	28.38	80.45
GDTR	**35.86**	**37.03**	**34.56**	**31.04**	96.18

Table 5: The PSNR values and average computation time (in seconds) of recovered videos by various methods on four YUV video sequences. Best results are marked bold.

Based on these results, we only need to utilize the recommended parameters $\gamma_{\mathcal{L}} = 3$ and , $\gamma_{\mathcal{E}} = 10^3$ to achieve a good denoising result.

Ablation Study: DTR is a tensor recovery model that incorporates structural priors of the low-rank component $\mathcal{L}$ and sparse component $\mathcal{E}$. Specifically, DTR adopts the regularizers $|||\mathcal{L}||_{T,*}$ and $\mathcal{E}||_{T,1}$ to replace the conventional counterparts $\|\mathcal{L}\|_{T,*}$ and $\|\mathcal{E}\|_1$ used in TRPCA. This motivates an investigation into whether either $\|\mathcal{L}\|_{T,*}$ or $\|\mathcal{E}\|_{T,1}$ alone can improve the recovery performance. To address this issue, we perform ablation studies on two variants: (a) DTR-L, which uses $\|\mathcal{L}\|_{T,*}$ together with $\|\mathcal{E}\|_1$; and (b) DTR-E, which uses $\|\mathcal{L}\|_*$ together with $\|\mathcal{E}\|_{T,1}$. Tab. 6 reports the PSNR results of these variants and DTR on the YUV video sequences. The above observations can be summarized as follows.

- Firstly, DTR-L and DTR-E achieve higher PSNR values than TRPCA on all three video sequences. This indicates that each individual regularization term, i.e., $\|\mathcal{L}\|_{T,*}$ and $\|\mathcal{E}\|_{T,1}$, can independently contribute to improving the recovery capability of TRPCA.

- Secondly, DTR is superior to both DTR-L and DTR-E, which indicates that it is meaningful to integrate $\|\mathcal{L}\|_{T,*}$ and $\|\mathcal{E}\|_{T,1}$ into a joint framework. This suggests that the two regularizers reinforce each other during optimization, rather than

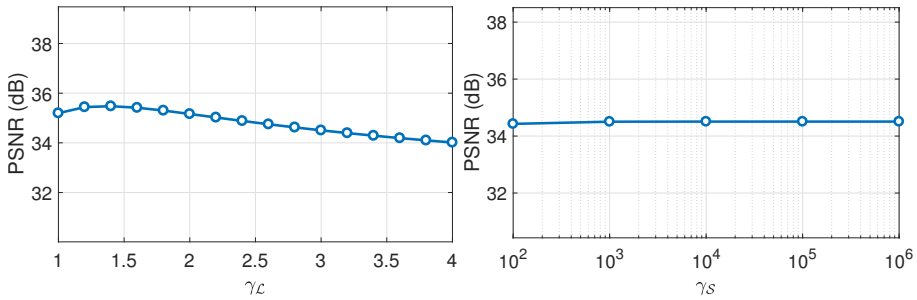


Figure 9: Changes of PSNR values of DTR with varing parameter $\gamma_{\mathcal{L}}$ and parameter $\gamma_{\mathcal{E}}$ for color image recovery on the example test image. (a) Changes of the PSNR values with varying parameter $\gamma_{\mathcal{L}}$ (fixed $\gamma_{\mathcal{E}} = 10^3$); (b) changes of the PSNR values with varying parameter $\gamma_{\mathcal{E}}$ (fixed $\gamma_{\mathcal{L}} = 3$).

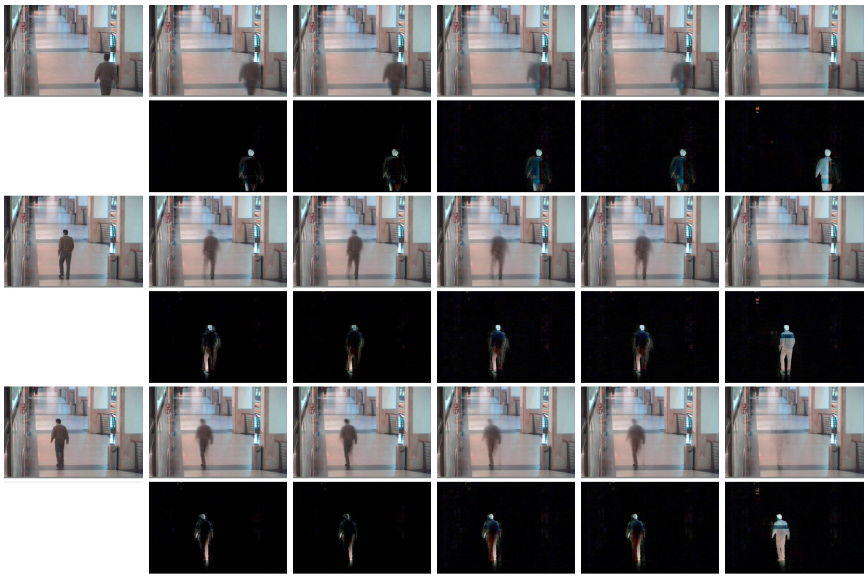
Methods	Regularization		Akiyo	Claire	Container
	$\|\mathcal{L}\|_{T,*}$	$\|\mathcal{E}\|_{T,1}$			
TRPCA			30.24	30.12	28.49
DTR-L	✓		30.80	30.59	29.16
DTR-E		✓	32.80	33.75	30.75
DTR	✓	✓	**33.57**	**35.12**	**31.85**

Table 6: Ablation Study. PSNR values of four methods with different regularization terms on the YUV video sequences. Best results are marked bold.

contributing independently as in DTR-L or DTR-E.

4.5. Application to Background Modeling

We investigate background modeling, which focuses on decomposing video data into foreground and background components. We conduct experiments on CAVIAR1 video sequence from Scene Background Initialization (SBI) database [39]. The CAVIAR1 sequence contains 610 frames with an original size of $256 \times 384 \times 3$. Each frame is resized to $64 \times 96 \times 3$ for subsequent processing. We evaluate DTR against several representative background modeling approaches, including RPCA [1], SNN [6], TRPCA [7], and ETRPCA [8]. For RPCA, the video sequence is reshaped into a matrix of size 18432×610. For SNN, TRPCA, ETRPCA, and DTR, the video data are reshaped into



(a) Original (b) RPCA (c) SNN (d) TRPCA (e) ETRPCA (f) DTR

Figure 10: Background modeling from videos. (a) Original frames; (b)-(f) low rank and sparse components obtained by RPCA, SNN, TRPCA, ETRPCA and our method DTR, respectively.

a tensor of size $6144\times3\times610$. The parameter settings of SNN follow the original paper [6]. The parameters of tensor based methods are also set to $\lambda = 1/\sqrt{d_3 \max(d_1, d_2)}$. Fig. 10 shows some original frames and the performance of RPCA, SNN, TRPCA, ETRPCA and DTR, respectively.The results demonstrate that DTR delivers more accurate recovery than the competing approaches. This can be attributed to its effective incorporation of the tube-sparse prior.

5. Conclusion

In this work, we develop a novel TRPCA method equipped with the newly defined tube-preserving tensor nuclear norm (T2NN) and tube-preserving tensor ℓ_1 norm (T2L1), referred to as DTR briefly. We simultaneously consider the tube sparse structure of the singular value tensor and sparse noise tensor. To achieve better recovery, the larger tubes are shrunk less while the smaller ones are shrunk more. We also develop an effective algorithm for DTR and establish its convergence property theoretically. Be-

sides, we extend DTR to a joint learning method of low-rankness and smoothness prior, which is coined as Gradient DTR (GDTR). Finally, extensive experiments validate the effectiveness of DTR and GDTR.

CRediT authorship contribution statement

Xiaowei Wang: Conceptualization, Methodology, Validation, Writing–original draft. **Libin Wang:** Formal analysis, Validation, Software. **Yulong Wang:** Supervision, Funding acquisition, Writing–review & editing. **Xinwei He:** Visualization, Investigation. **Yutao Hu:** Data curation, Validation. **Han Li:** Writing–review & editing. **Yuan Yan Tang:** Supervision, Funding acquisition, Writing–review & editing.

References

[1] E. Candès, X. Li, Y. Ma, J. Wright, Robust principal component analysis?, J. ACM 58 (3) (2011) 1–37.

[2] L. Wu, J. Wang, W.-S. Zheng, G. Shi, Double nonconvex tensor robust kernel principal component analysis and its visual applications, IEEE Trans. Image Process. 35 (2026) 1711–1726.

[3] Q. Liu, H. Ge, X. Su, Low-rank tensor recovery via jointing the non-convex regularization and deep prior, Neurocomputing 648 (2025) 130610.

[4] M. Kilmer, C. Martin, Factorization strategies for third-order tensors, Linear Algebra Appl. 435 (3) (2011) 641–658.

[5] T. G. Kolda, B. W. Bader, Tensor decompositions and applications, SIAM Rev. 51 (3) (2009) 455–500.

[6] J. Liu, P. Musialski, P. Wonka, J. Ye, Tensor completion for estimating missing values in visual data, IEEE Trans. Pattern Anal. Mach. Intell. 35 (1) (2013) 208–220.

[7] C. Lu, J. Feng, Y. Chen, W. Liu, Z. Lin, S. Yan, Tensor robust principal component analysis with a new tensor nuclear norm, IEEE Trans. Pattern Anal. Mach. Intell. 42 (4) (2020) 925–938.

[8] Q. Gao, P. Zhang, W. Xia, D. Xie, X. Gao, D. Tao, Enhanced tensor rpca and its application, IEEE Trans. Pattern Anal. Mach. Intell. 43 (6) (2021) 2133–2140.

[9] H. Wang, F. Zhang, J. Wang, T. Huang, J. Huang, X. Liu, Generalized nonconvex approach for low-tubal-rank tensor recovery, IEEE Trans. Neural Netw. Learn. Syst. 33 (8) (2022) 3305–3319.

[10] J. Wang, J. Hou, Y. C. Eldar, Tensor robust principal component analysis from multilevel quantized observations, IEEE Trans. Inf. Theory 69 (1) (2023) 383–406.

[11] X. Luo, W. Ma, W. Wang, Y. Zheng, J. Wang, Low-rank high-order tensor recovery via joint transformed tensor nuclear norm and total variation regularization, Neurocomputing 624 (2025) 129266.

[12] L. Yang, H. Yang, T. X. Wu, Y. Wang, T. Yang, Y. Y. Tang, Tensor singular value-preserving norm for robust visual data recovery, Knowl.-Based Syst. 341 (2026) 115831.

[13] Y. Wu, J. Sun, J. Yin, Tensor nonconvex unified prior for tensor recovery, Information Sciences 680 (2024) 121176.

[14] F. Zhang, J. Wang, W. Wang, C. Xu, Low-tubal-rank plus sparse tensor recovery with prior subspace information, IEEE Trans. Pattern Anal. Mach. Intell. 43 (10) (2021) 3492–3507.

[15] Y. Chen, S. Wang, Y. Zhou, Tensor nuclear norm-based low-rank approximation with total variation regularization, IEEE J. Sel. Topics Signal Process. 12 (6) (2018) 1364–1377.

[16] C.-Y. Ko, K. Batselier, L. Daniel, W. Yu, N. Wong, Fast and accurate tensor completion with total variation regularized tensor trains, IEEE Trans. Image Process. 29 (2020) 6918–6931.

[17] J. Peng, H. Wang, X. Cao, X. Liu, X. Rui, D. Meng, Fast noise removal in hyperspectral images via representative coefficient total variation, IEEE Trans. Geosci. Remote Sens. 60 (2022) 1–17.

[18] J. Peng, Y. Wang, H. Zhang, J. Wang, D. Meng, Exact decomposition of joint low rankness and local smoothness plus sparse matrices, IEEE Trans. Pattern Anal. Mach. Intell. 45 (5) (2022) 5766–5781.

[19] K. Huang, W. Kong, M. Zhou, W. Qin, F. Zhang, J. Wang, Enhanced low-rank tensor recovery fusing reweighted tensor correlated total variation regularization for image denoising, J. Sci. Comput. 99 (3) (2024) 69.

[20] Z. Cheng, R. Zhang, G. Zhang, Y. Xu, X. Ji, W. Zhou, Low-rank tensor recovery via variational schatten-p quasi-norm and jacobian regularization, Neurocomputing 699 (2026) 134312.

[21] H. Wang, J. Peng, W. Qin, J. Wang, D. Meng, Guaranteed tensor recovery fused low-rankness and smoothness, IEEE Trans. Pattern Anal. Mach. Intell. 45 (9) (2023) 10990–11007.

[22] L. Yuan, C. Li, J. Cao, Q. Zhao, Randomized tensor ring decomposition and its application to large-scale data reconstruction, in: Proc. ICASSP, 2019, pp. 2127–2131.

[23] T. Shi, M. Ruth, A. Townsend, Parallel algorithms for computing the tensor-train decomposition, SIAM J. Sci. Comput. 45 (3) (2023) C101–C130.

[24] J. Zhang, A. K. Saibaba, M. E. Kilmer, S. Aeron, A randomized tensor singular value decomposition based on the t-product, Numer. Linear Algebra Appl. 25 (5) (2018) e2179.

[25] W. Qin, H. Wang, F. Zhang, W. Ma, J. Wang, T. Huang, Nonconvex robust high-order tensor completion using randomized low-rank approximation, IEEE Trans. Image Process. 33 (2024) 2835–2850.

[26] S. Boyd, N. Parikh, E. Chu, B. Peleato, J. Eckstein, Distributed optimization and statistical learning via the alternating direction method of multipliers, Found. Trends Mach. Learn. 3 (2011) 1–122.

[27] Q. Gao, W. Xia, Z. Wan, D. Xie, P. Zhang, Tensor-svd based graph learning for multi-view subspace clustering, in: Proc. AAAI, Vol. 34, 2020, pp. 3930–3937.

[28] Z. Zhang, G. Ely, S. Aeron, N. Hao, M. Kilmer, Novel methods for multilinear data completion and de-noising based on tensor-svd, in: Proc. CVPR, 2014, pp. 3842–3849.

[29] F. Zhang, H. Wang, W. Qin, X. Zhao, J. Wang, Generalized nonconvex regularization for tensor rpca and its applications in visual inpainting, Appl. Intell. 53 (20) (2023) 23124–23146.

[30] R. He, W.-S. Zheng, B.-G. Hu, Maximum correntropy criterion for robust face recognition, IEEE Trans. Pattern Anal. Mach. Intell. 33 (8) (2011) 1561–1576.

[31] R. He, W.-S. Zheng, T. Tan, Z. Sun, Half-quadratic-based iterative minimization for robust sparse representation, IEEE Trans. Pattern Anal. Mach. Intell. 36 (2) (2014) 261–275.

[32] P. Arbelaez, M. Maire, C. Fowlkes, J. Malik, Contour detection and hierarchical image segmentation, IEEE Trans. Pattern Anal. Mach. Intell. 33 (5) (2011) 898–916.

[33] Y. Zhou, Y.-M. Cheung, Bayesian low-tubal-rank robust tensor factorization with multi-rank determination, IEEE Trans. Pattern Anal. Mach. Intell. 43 (1) (2021) 62–76.

[34] M. Yang, Q. Luo, W. Li, M. Xiao, Nonconvex 3d array image data recovery and pattern recognition under tensor framework, Pattern Recognit. 122 (2022) 108311.

[35] L. Chen, L. Ge, X. Jiang, H. Li, M. Haardt, Slim is better: Transform-based tensor robust principal component analysis, IEEE Trans. Signal Process. 73 (2025) 2320–2335.

[36] L. Chen, L. Ge, X. Jiang, H. Li, Boosting rpca by prior subspace, IEEE Trans. Signal Process. 73 (2025) 2170–2186.

[37] Z. Wang, A. C. Bovik, H. R. Sheikh, E. P. Simoncelli, Image quality assessment: from error visibility to structural similarity, IEEE Transactions on Image Processing 13 (4) (2004) 600–612.

[38] Q. Shi, Y.-M. Cheung, J. Lou, Robust tensor svd and recovery with rank estimation, IEEE Trans. Cybern. 52 (10) (2022) 10667–10682.

[39] P. A. Maddalena. Lucia, Towards benchmarking scene background initialization, in: ICIAP, 2015, pp. 469–476.

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☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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